

Permutations

Counting & Arrangements Worksheet · Grade 9–12

Name: _____

Date: _____

Learning Objectives

- Apply the permutation formula $nPr = n! / (n - r)!$ to evaluate expressions and solve real-world problems
- Identify the values of n and r in a permutation context and understand that n must be greater than r
- Calculate the number of distinct arrangements of words with repeated letters using the formula $n! / (r_1! \cdot r_2! \cdot \dots)$

Problems

1. Evaluate the permutation expression.

$${}_5P_2$$

2. Evaluate the permutation expression.

$${}_6P_3$$

3. Evaluate the permutation expression.

$${}_8P_2$$

4. A club has 8 members. In how many ways can they choose a President, Vice-President, and Secretary (all different people)?

$${}_8P_3$$

5. Six runners are in a race. How many different ways can the gold, silver, and bronze medals be awarded?

$${}_6P_3$$

6. How many 4-letter arrangements can be made from the letters in the word PENCIL (no letter repeated)?

$${}_6P_4$$



7. Find the number of distinct arrangements of all the letters in the word LEVEL.

$$\frac{5!}{2! \cdot 2!}$$

8. Find the number of distinct arrangements of all the letters in the word MISSISSIPPI.

$$\frac{11!}{4! \cdot 4! \cdot 2!}$$

9. A photographer must arrange 4 people from a group of 10 in a single row for a photo. How many different arrangements are possible?

$$10P4$$

10. A security system requires a 3-digit PIN using digits from 1 through 9 with no digit repeated. How many possible PINs are there? Then find how many of these PINs start with the digit 5.

$$9P3$$



Permutations — Answer Key

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Answer Key

1. Answer: 20

- Use the formula: $nPr = n! / (n - r)!$
 - $5P2 = 5! / (5 - 2)! = 5! / 3!$
 - $= (5 \times 4 \times 3!) / 3! = 5 \times 4 = 20$
-

2. Answer: 120

- Use the formula: $nPr = n! / (n - r)!$
 - $6P3 = 6! / (6 - 3)! = 6! / 3!$
 - $= 6 \times 5 \times 4 \times 3! / 3! = 6 \times 5 \times 4 = 120$
-

3. Answer: 56

- Use the formula: $nPr = n! / (n - r)!$
 - $8P2 = 8! / (8 - 2)! = 8! / 6!$
 - $= 8 \times 7 \times 6! / 6! = 8 \times 7 = 56$
-

4. Answer: 336

- Order matters (different titles), so use permutation: $n = 8, r = 3$
 - $8P3 = 8! / (8 - 3)! = 8! / 5!$
 - $= 8 \times 7 \times 6 \times 5! / 5! = 8 \times 7 \times 6 = 336$
-

5. Answer: 120

- Order matters (1st, 2nd, 3rd places are distinct), so use permutation: $n = 6, r = 3$
 - $6P3 = 6! / (6 - 3)! = 6! / 3!$
 - $= 6 \times 5 \times 4 \times 3! / 3! = 6 \times 5 \times 4 = 120$
-

6. Answer: 360

- PENCIL has 6 distinct letters, and we arrange 4 at a time: $n = 6, r = 4$
 - $6P4 = 6! / (6 - 4)! = 6! / 2!$
 - $= 6 \times 5 \times 4 \times 3 \times 2! / 2! = 6 \times 5 \times 4 \times 3 = 360$
-

7. Answer: 30

- LEVEL has 5 letters: L appears 2 times, E appears 2 times, V appears 1 time
 - Use the repeated-letters formula: $n! / (r_1! \cdot r_2! \cdot \dots)$
 - $= 5! / (2! \times 2!) = 120 / (2 \times 2) = 120 / 4 = 30$
-

8. Answer: 34,650

- MISSISSIPPI has 11 letters: M=1, I=4, S=4, P=2
 - Use the repeated-letters formula: $11! / (1! \times 4! \times 4! \times 2!)$
 - $= 39,916,800 / (1 \times 24 \times 24 \times 2) = 39,916,800 / 1,152 = 34,650$
-



9. Answer: 5,040

- Order matters (positions in a row are distinct): $n = 10, r = 4$
 - ${}_{10}P_4 = 10! / (10 - 4)! = 10! / 6!$
 - $= 10 \times 9 \times 8 \times 7 \times 6! / 6! = 10 \times 9 \times 8 \times 7 = 5,040$
-

10. Answer: Total PINs: 504; PINs starting with 5: 56

- Part 1 — Total PINs: $n = 9, r = 3 \rightarrow {}_9P_3 = 9! / 6! = 9 \times 8 \times 7 = 504$
 - Part 2 — PINs starting with 5: Fix digit 5 in the first position; choose remaining 2 digits from the 8 remaining digits in order
 - ${}_8P_2 = 8! / 6! = 8 \times 7 = 56$
 - So there are 504 total PINs and 56 that start with the digit 5
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